

Week 8 - Some linear equations

Definition: Define a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ to be *linear* if both:

1. For all $x, y \in \mathbb{R}^n$, we have $f(x + y) = f(x) + f(y)$.
 2. For all $c \in \mathbb{R}$ and $x \in \mathbb{R}^n$, we have $f(cx) = cf(x)$.
- 1) Is the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by $f(x, y, z) = 12x - 2z$ linear or not? Prove your answer.

2) Is the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = 12x - 2$ linear or not? Prove your answer.

3) Is the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $f(x, y) = (2x, 8x)$ linear or not? Prove your answer.

4) Is the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = 12x - 2y^2$ linear or not? Prove your answer.

5) Is the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 12$ linear or not? Prove your answer.

6) Is the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = 0$ linear or not? Prove your answer.

7) Suppose $ax + by = 4$ is a linear equation.

(a) Prove that there is a linear function $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that the preimage $L^{-1}(4)$ is the set of pairs $(x, y) \in \mathbb{R}^2$ that satisfy the equation.

(b) Prove that there exists a function $L' : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that the preimage $L'^{-1}(0)$ is the set of pairs $(x, y) \in \mathbb{R}^2$ that satisfy the equation.

(c) Does there exist a *linear* function $L' : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that the preimage $L'^{-1}(0)$ is the set of pairs $(x, y) \in \mathbb{R}^2$ that satisfy the equation? Prove your answer.

10) Suppose we are given a system of linear equations,

$$2x - y = 0$$

$$-x + 9y = 5.$$

Show that there exists a linear function $L : \mathbb{R}^2 \rightarrow \mathbb{R}$, such that the set of solutions to the system is exactly the preimage, $L^{-1}((0, 5))$.

11) Given any linear function $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$, do you think it is true that there is always a system of linear equations whose solution set is exactly $L^{-1}(\mathbf{0})$, where $\mathbf{0}$ is the origin (zero vector) in $\mathbb{R}^m$? Explain why. *Bonus:* Try to prove your answer.